Hardware Prototyping Tips

ME400: Capstone Design I

2026, Spring

Engineering for Value Creation

KAIST

Contents

This article mainly focus on Practical/Experimental knowledges.

For **Mechanics**,

- Design Strategies
- Material Properties: *PLA, Wood, Steel, Alu, etc.*
- Machine Tool Properties: *3D Printer, Laser cutter, etc.*
- Attachment & Tolerance: *Bolts, Taps, etc.*

For **Electronics,**

- Electronics Properties: *DC motor, RC Servo, LiPo, Li-ion battery, etc.*
- Soldering Tips

+ Model names for the parts

Design Strategies

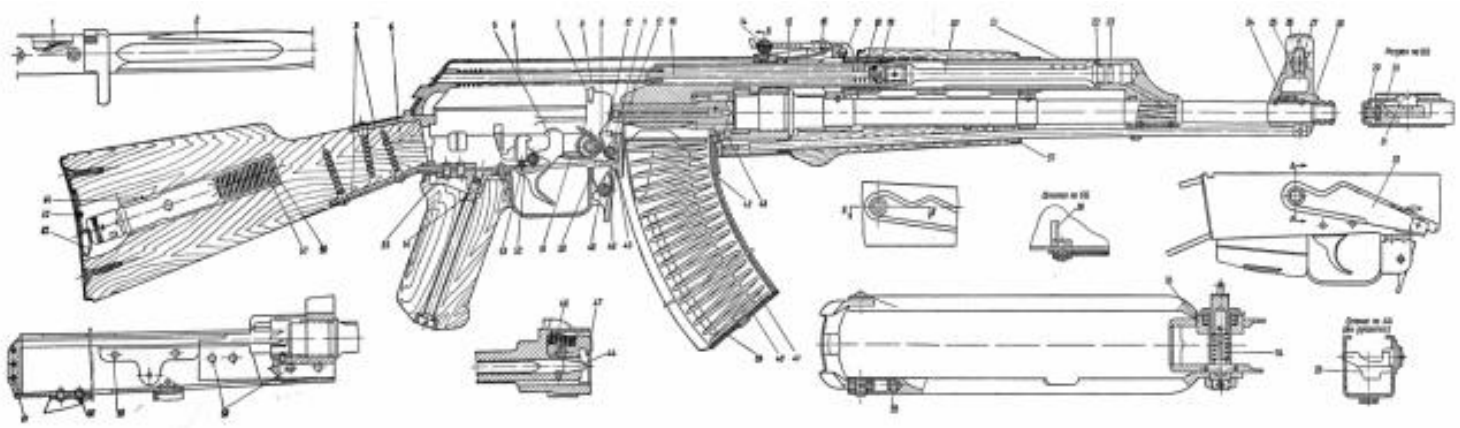
Design Strategies

These are 5 design strategies which are simple, but very important.

- Simple is the Best.
- Be conservative.
- Understand Materials & Tools.
- Consider Tolerance.
- Do it Faster.

Design Strategies: Simple is the Best.

- The more mechanical parts, the higher the chance of unpredictable problems, and the lower the structural stiffness & strength to weight ratio.
- In addition, maintenance becomes difficult and the unit price rises.
- Always think if there is a **simpler way** that can accomplish the objective.



Ex) Success of simplicity, Design of AK47 series.

Design Strategies: Be coservative.

- The process of completing the mission requires creative ideas. However, some people are obsessed with creativity itself and invest an excessive amount of time in coming up with **something new.**
- If you make a challenging and new attempt, you should think about why there were no such attempts before, what problems are expected and how to solve them.
- You need concrete reasons and grounds for success of your new attempts.

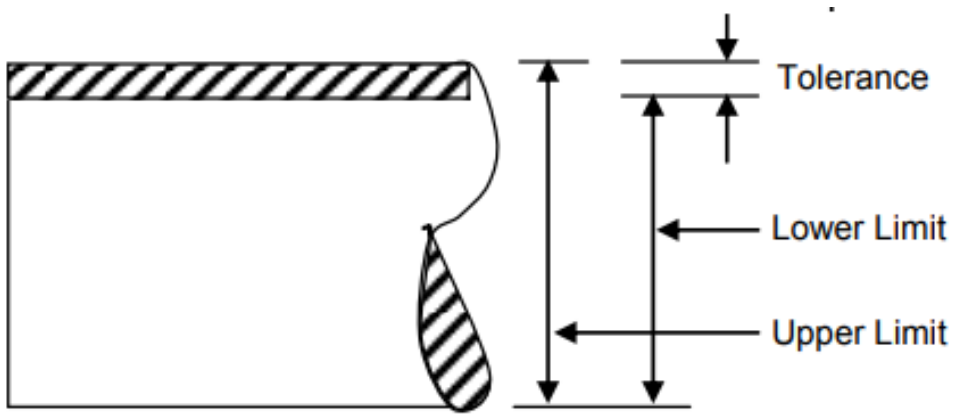
Ex) A useless invention.

Design Strategies: Understand Materials & Tools.

- Even for prototyping, many kinds of materials can be used. Therefore, when designing, physical properties such as *stiffness, hardness, brittleness, strength, price, and mass* of each material should be considered.
- In addition, even for the same material, the time and effort required and the dimensional accuracy of the results are different depending on the equipment and method, so this should also be carefully considered.

Design Strategies: Consider Tolerance.

- All materials are not rigid, and all processing make *tolerance*. Many students do not consider *tolerances*, only consider *nominal dimensions*. This causes a non-assembly or loose assembly state with a very high probability.
- Therefore, smooth and rigid assembly between parts is possible when it is designed in consideration of tolerances according to materials and processing methods.



Design Strategies: Do it Faster.

- The core of prototyping is to produce results in limited time. In the process of prototyping, many unexpected problems such as failure of parts, procedures, and design can occur, so design iteration must proceed at a rapid pace in order to save time for them.
- No matter how good a designer is, at least one revisions is unavoidable. Starting building after preparing and planning all problems in advance is inefficient, especially in terms of time, and is not possible.
- So try, build, and experiment ASAP.

Material Properties

Material Properties

The materials below are suitable for prototyping because they are easy to use in KAIST campus.

Their characteristics and uses are explained in this document.

- Structural Materials(general):

  :PLA*(Poly Lactic Acid),* Acrylic Board, MDF*(Medium Density Fiberboard),* Aluminum profile

PLA

Material Properties: PLA

- PLA(*Poly Lactic Acid)* is plastic used by general *FDM* 3D Printers.
- Pros:

- With a 3D printer, it is easy to process relatively complex shapes **easily.**
- Compared to this processability, the **mechanical properties** are also good.

- Cons:

- Because it is weak against **heat**, it melts or bends when exposed to high temperatures over **60°C**.
- **Sanding** is hard due to weakness at heat.

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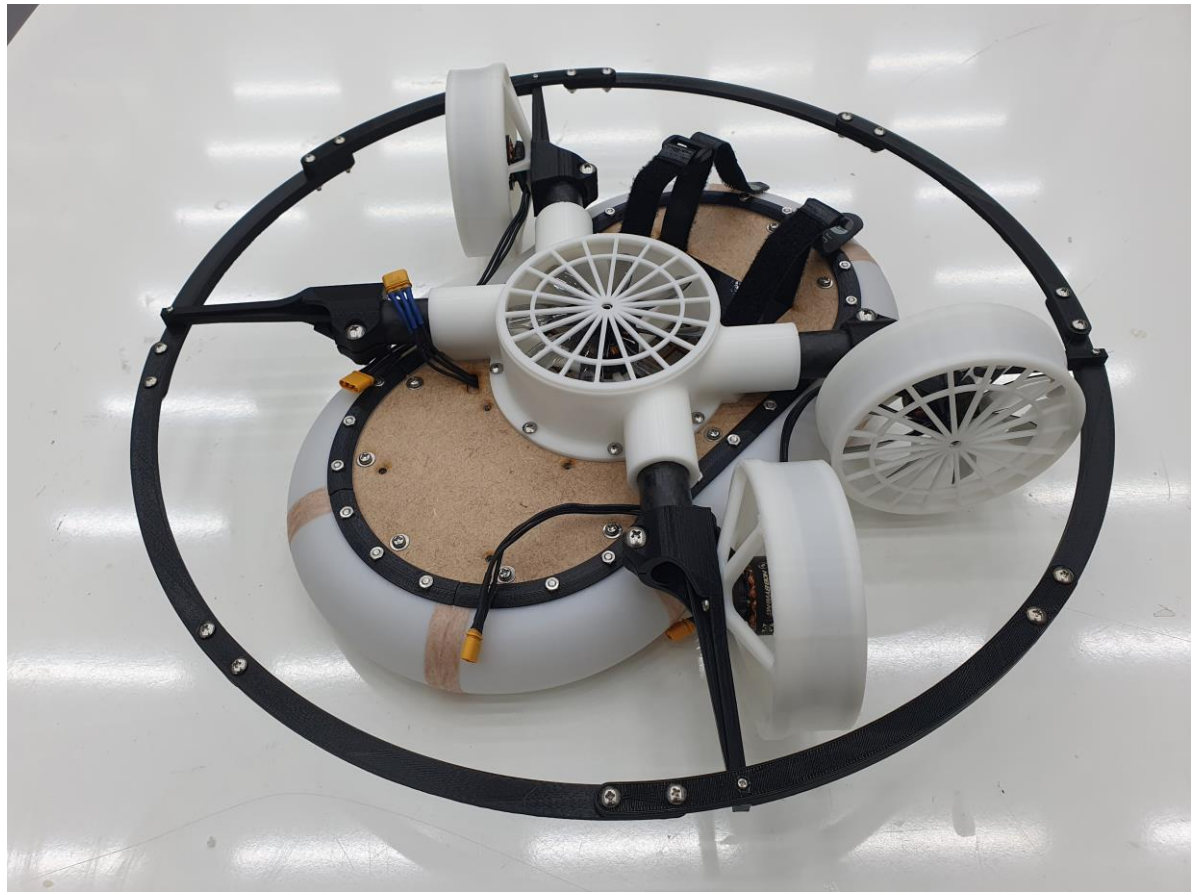
Material Properties: PLA

- **Usage**: PLA can be used with FDM 3D printers to make most of the structural parts that cannot be made with 2D plates and there are no ready-made products.



- As you can see in the example on the left, most 3D shaped parts can be made from PLA using a 3D printer.

- Skirt of this Hovercraft is also made of PLA using a FDM 3D printer, with 0.8mm thickness.

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Material Properties: PLA

- The followings are the mechanical properties of PLA.

Density: **1.24g/cm^3**

Tensile/Yield Strength: **50~60MPa (20°C)**

Young's modulus: 3600Mpa

Elongation: 4~9%

Transition temperature: 55~65°C

Melting point: 140~160°C

For more information, visit URL below.

https://www.matweb.com/search/DataSheet.aspx?MatGUID=ab96a4c0655c4018a8785ac4031b9278

Acrylic Board

Material Properties: Acrylic Board

- Acrylic board is the plastic board used by general laser cutters.
- Pros:

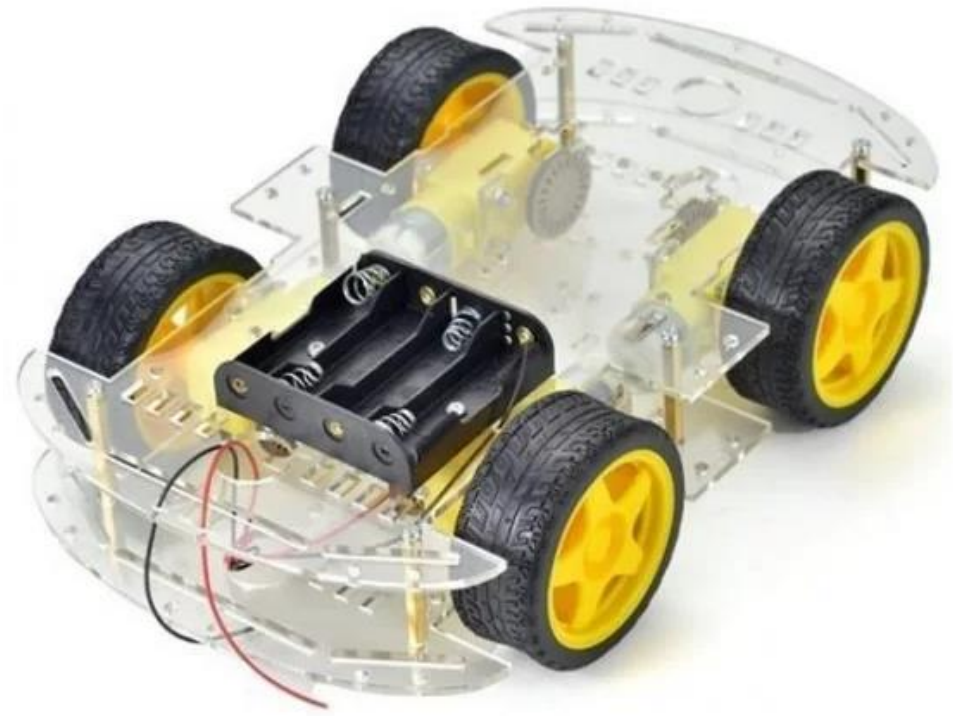
- There are colorless and transparent varieties.
- Easy to process with a laser cutter.
- Hardness, Stiffness is higher than other plastics.

- Cons:

- Heavy, compared to wood.
- Brittleness is high.
- Only can use on 2D designs.

Material Properties: Acrylic Board

- **Usage**: Acrylic board can be used as frame of small products(~1kg), In the case of no impact force.



- As the acrylic board is highly brittle, it should not be used where impact is applied or vibration is severe.
- When making hardware over about 1kg, the frame should be made of plastic, aluminum, or carbon fiber that has better mechanical performance than the acrylic board.

Material Properties: Acrylic Board

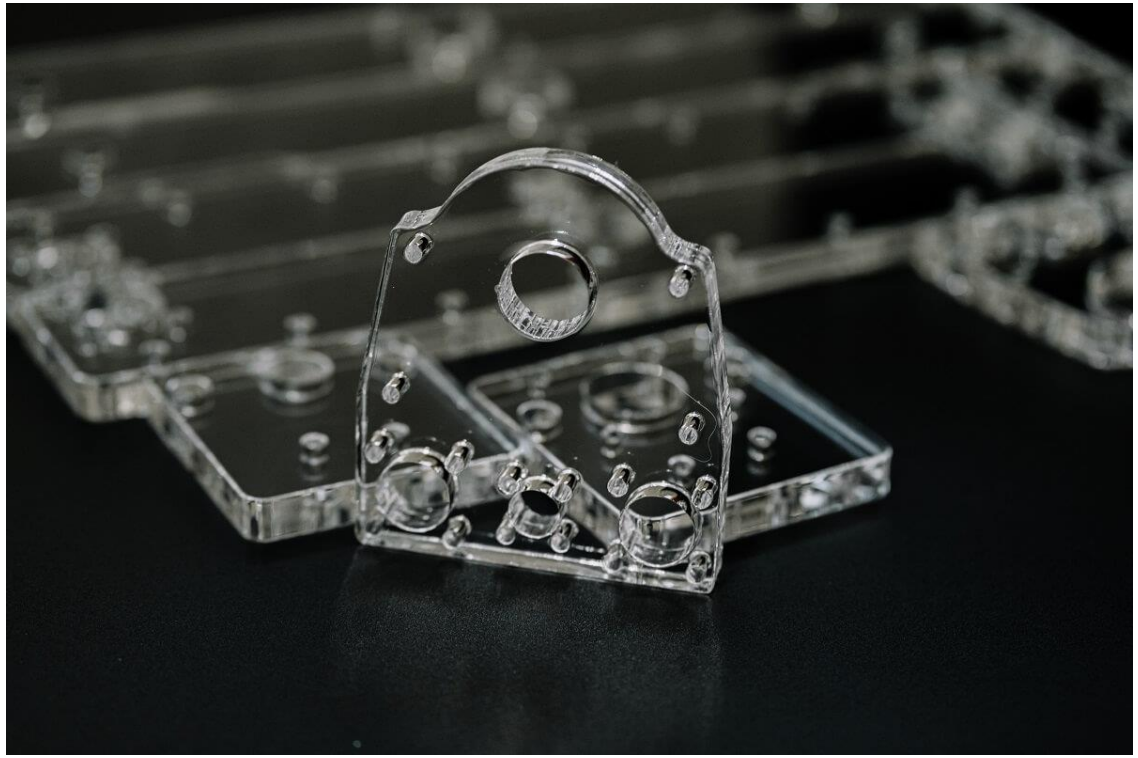
- The followings are the mechanical properties of Acrylic board.

Density: **1.18g/cm^3**

Yield Strength: **64.8~83.4MPa**

Young's modulus: 2.67~3.30GPa

Elongation: 3.0~6.4%



For more information, visit URL below.

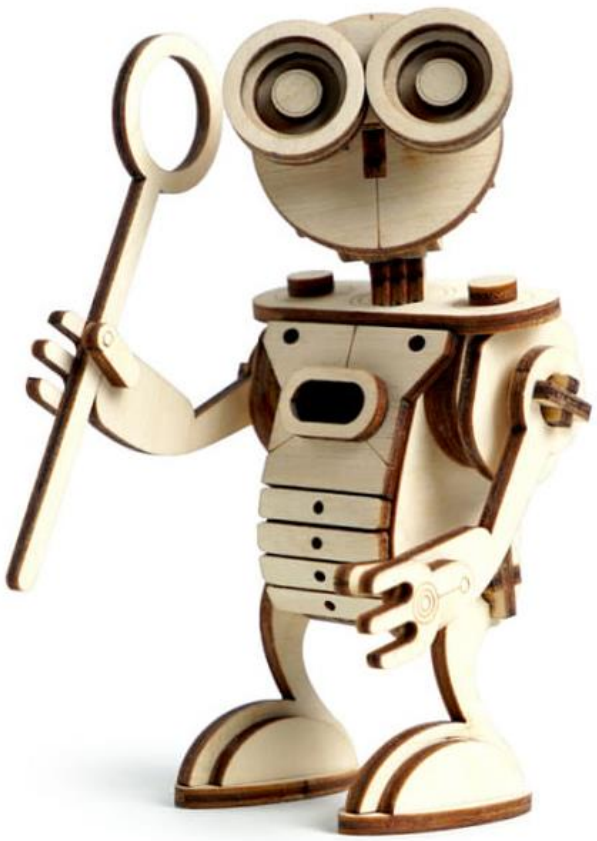
https://www.matweb.com/search/datasheet.aspx?bassnum=O1303

MDF

Material Properties: MDF

- MDF*(Medium Density Fiberboard)* is the wood board used by general laser cutters.
- Pros:

- Light weight(~0.7g/cm^3).
- Strong against heat compare to PLA.
- Easy to process with a laser cutter.
- Cheap.



- Cons:

- Strength, Stiffness and Hardness is low.
- Only use on 2D designs.

Material Properties: MDF

- **Usage**: MDF can be used as a frame of small products(~1kg), when the structural properties (Strength, Stiffness, Hardness) are not that needed.

- The lighter the *hull* of a Hovercraft, the better, and because it does not require high strength, the *hull* was made of 3mm thickness MDF.

Material Properties: MDF

- The followings are the mechanical properties of MDF.

Density: **0.6~0.8g/cm^3**

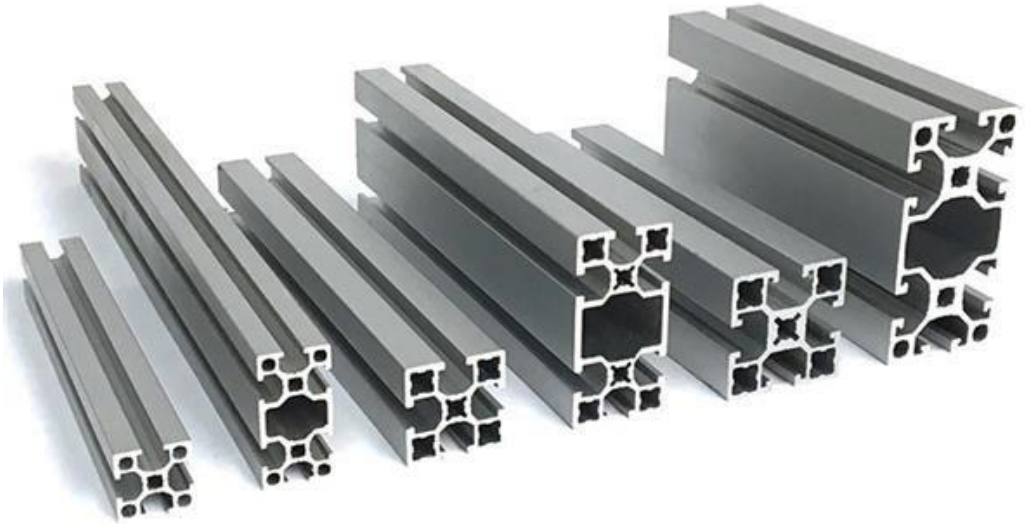
Ultimate tensile Strength: **18MPa**

Young's modulus: 4GPa

Elongation: 0.5%

For more information, visit URL below.

https://www.makeitfrom.com/material-properties/Medium-Density-Fiberboard-MDF

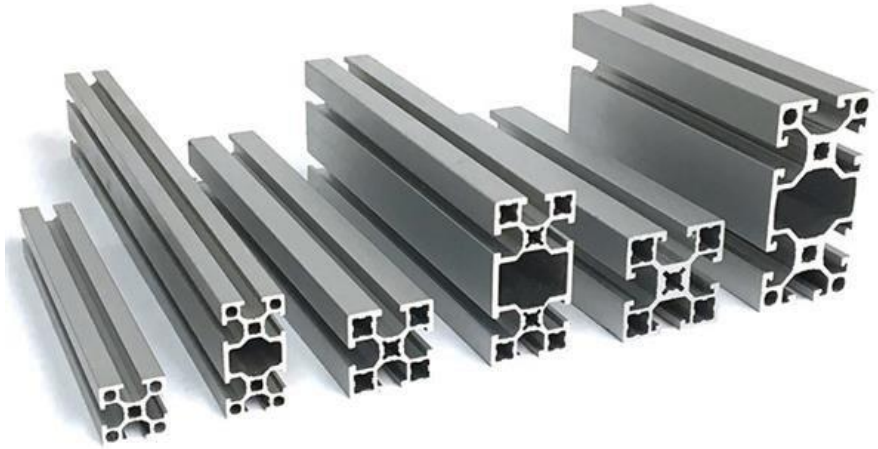


Aluminum Profile

Material Properties: Aluminum Profile

- An aluminum profile is a frame for structures.
- Pros:

- Pros of Aluminum, better mechanical properties than plastic or wood.
- Easy to make structures.



- Cons:

- *High speed cutter* is needed to process.
- Heavy compared to plastic, wood.

Machine Tool Properties

This document briefly describes the purpose and characteristics of each device.
For detailed explanation, Search websites or books recommended in this document.

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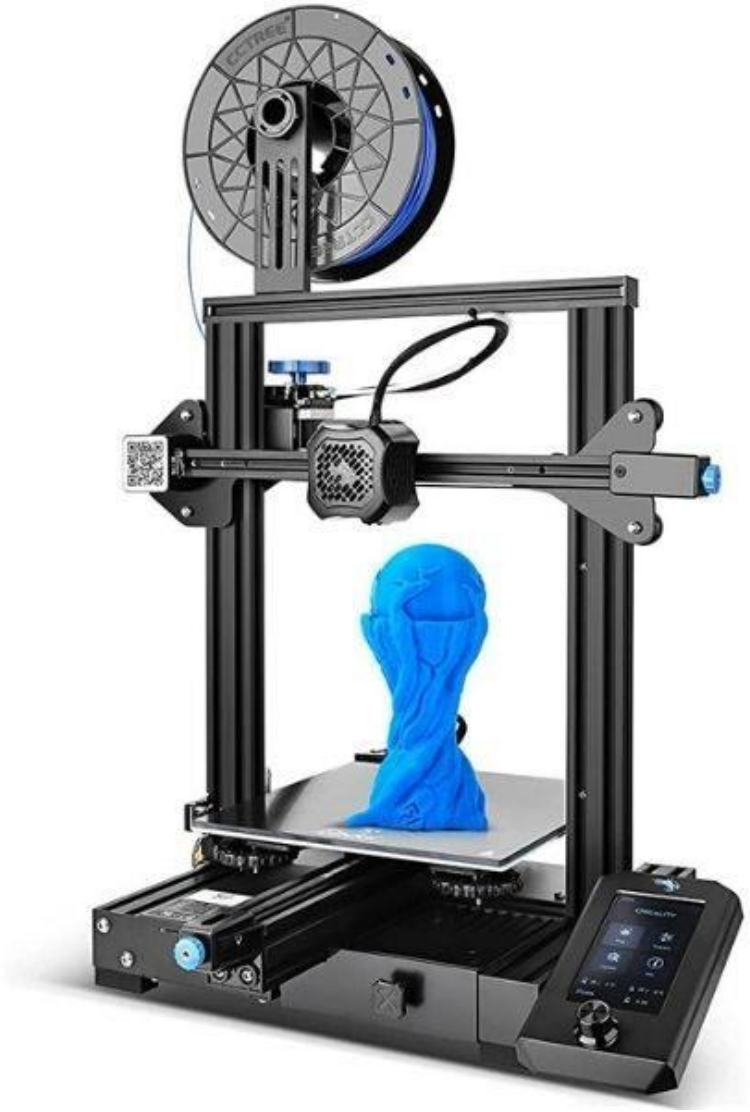

FDM type 3D Printer

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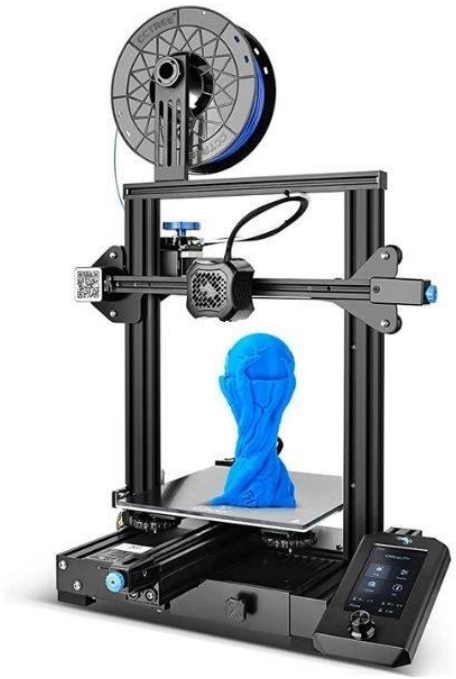
Machine Tool Properties: FDM 3D Printer



- **Prerequisite:**

- Skill on 3D Modeling tool(SolidWorks, Fusion360, etc.)

- **Books for 3D printing:**

- 3D 프린터 101 : 기초부터 활용까지, 3D 프린터의 모든 것 (안상준, 한빛미디어)

- **Videos/Websites for 3D printing:**

- https://www.notion.so/3D-Printer-License-830fb3b5786e4a01922b2951e0110b34
- https://www.youtube.com/watch?v=nb-Bzf4nQdE&list=PLDJMid0lOOYnkcFhz6rfQ6Uj8x7meNJJx
- https://www.youtube.com/watch?v=3LBTkLsjHGQ

'Ender 3 Pro' is very widely used FDM 3D printer due to its low price***(200,000₩~300,000₩).***

In the case of the FDM type, the accuracy performance of a 3D printer has little correlation with price.

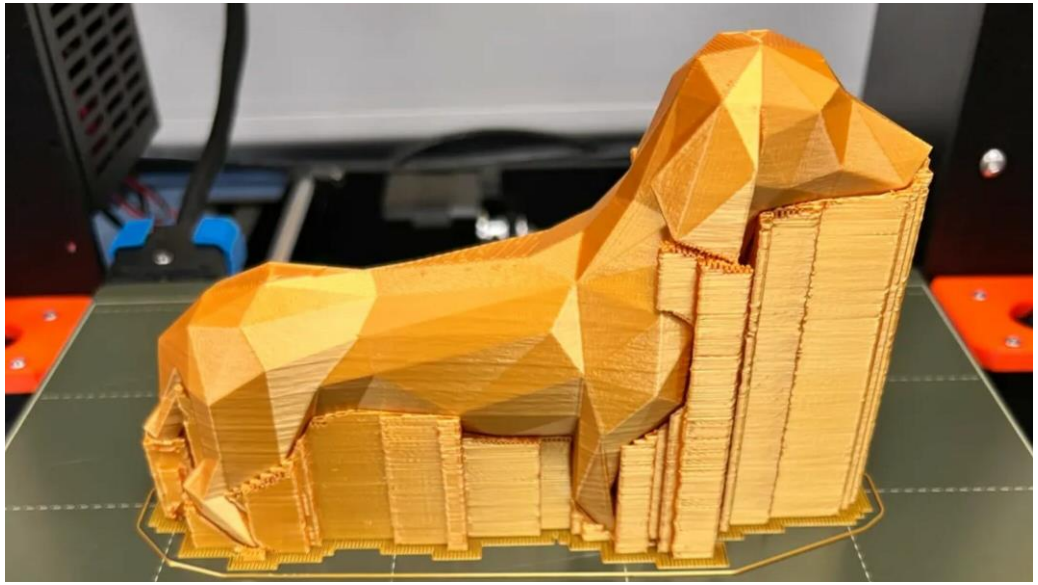
(Expensive printers are easier to maintain.)

Machine Tool Properties: FDM 3D Printer

- 3D Printing **Design** Tips:

No Support:

- When you design 3D printing parts, design so that *Support* is NOT created as much as possible. This is because, due to the sparse structure of the *support*, the printing accuracy of the part stacked on it is considerably lowered(~1mm).

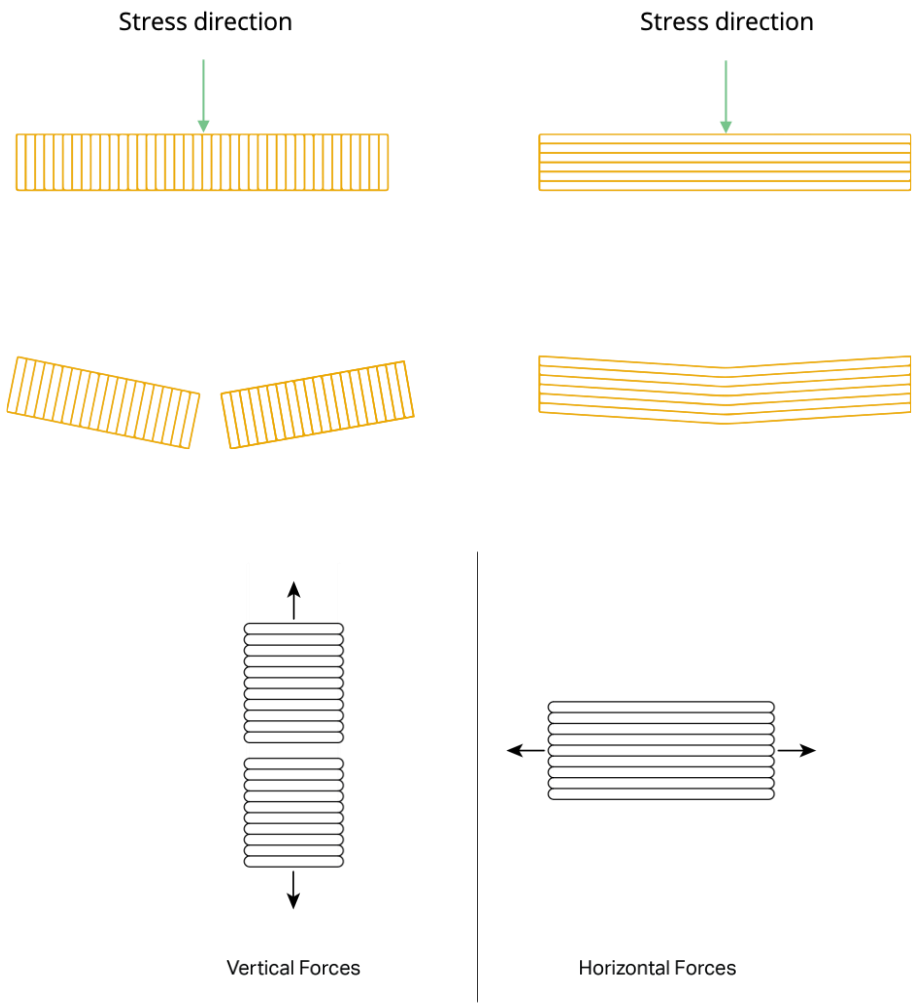


Machine Tool Properties: FDM 3D Printer

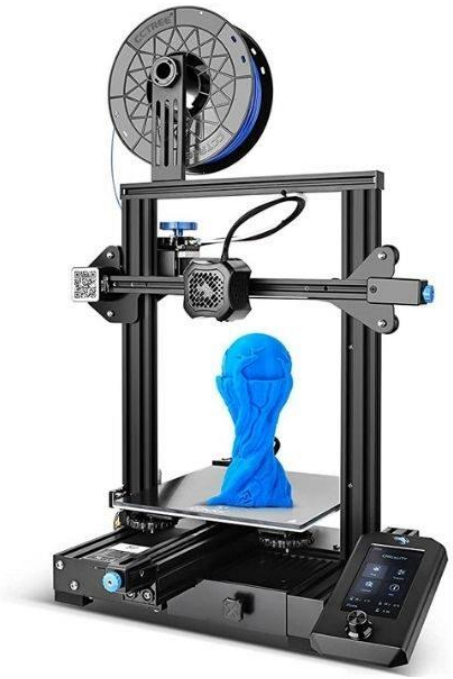
- 3D Printing **Design** Tips:

Anisotropic strength:

- Since FDM 3D printers generally laminate materials in the vertical axis (z axis) direction, the horizontal strength and vertical strength are different. In general, in the case of PLA parts printed with an FDM 3D printer, the horizontal strength is about **50Mpa** and the vertical strength is about **20~30Mpa** at room temperature. Therefore, when designing a part to be 3D printing parts, it is necessary to consider the direction of the load that the part will mainly receive when printing.



Machine Tool Properties: FDM 3D Printer



- 3D **Slicer Program** Tips:

Infill:

- No strength is needed: **10~20%**
- Strength is needed: Thin(~10mm): **80%~100%**, Thick: **40%~70%**
- Too high Infill with Thick parts cause *Warping* 1).

Layer height:

- Most, 0.20mm(Default). Values lower than 0.20mm show almost same strength.

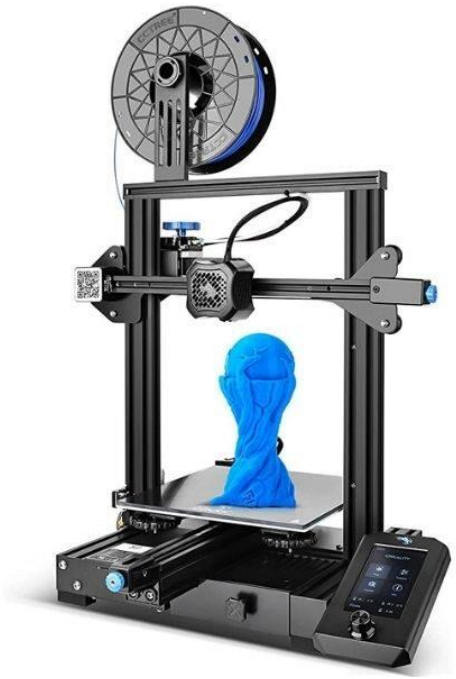
  When smooth surface is needed, about ~0.12mm.

Wall thickness:

- Most, 0.80mm(Default).
- When part contains Tap hole(Bolt hole) under load, make it higher(1.6mm~).

1) https://www.simplify3d.com/resources/print-quality-troubleshooting/warping/

Machine Tool Properties: FDM 3D Printer



- 3D **Slicer Program** Tips:

Build plate adhesion:

- Raft:

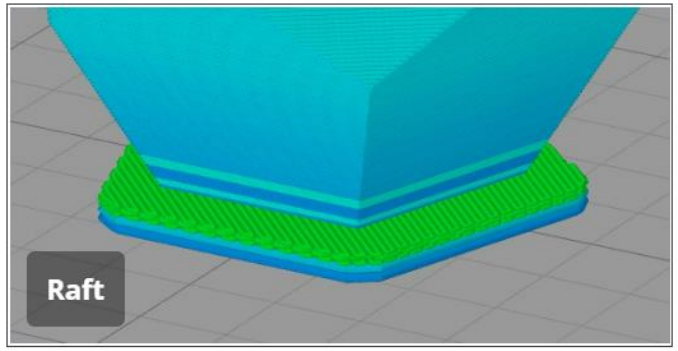
  Pros: Easy to tear off from *build plate*, OK even with Bad(tilted) *leveling.*

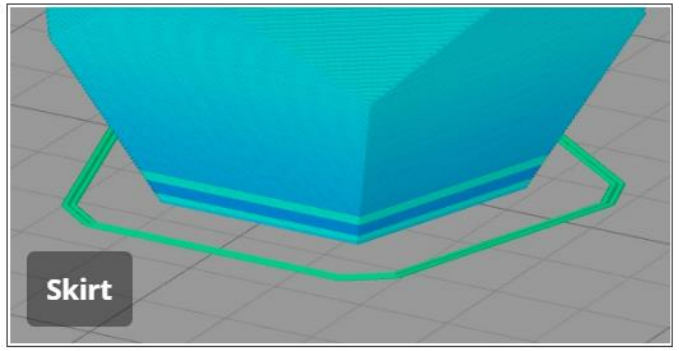
  Cons: Uneven bottom surface quality, longer print time.

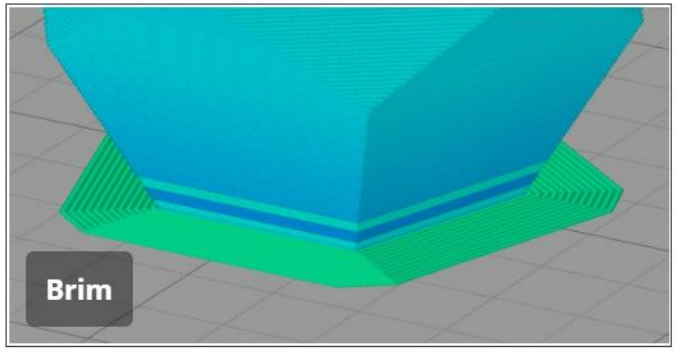
- Skirt, Brim:

  Pros: Higher bottom surface quality, shorter print time.

  Cons: Relatively precise *leveling* is needed, harder to tear off from build plate.







Laser Cutter

Machine Tool Properties: Laser Cutter

- **Characteristics:**

- **Pros:**

  Much higher speed(~5min) than FDM 3D printers(1 hour~1 day).

  Better quality for large, thin sheet shape parts compared to 3D printer.

- **Cons:**

  Only for 2D design, sheet material.

  Defects of the usable materials(MDF, Acrylic).

- **Processing Accuracy:**

  ~$\pm$0.200mm. One should consider the diameter of the laser beam.

Machine Tool Properties: Laser Cutter

- **Prerequisite**

- Little knowledge on 2D Modeling tool(AutoCAD, etc.)

- **Videos/Websites for Laser cutting:**

- https://www.notion.so/Laser-Cutter-License-54627ecf257d41b78f139eaba61afc8d
- https://www.youtube.com/watch?v=yUjgRasURnE&t=154s
- https://ckmakers.com/80/?q=YToxOntzOjEyOiJrZXl3b3JkX3R5cGUiO3M6MzoiYWxsIjt9&bmode=view&idx=3291906&t=board

- **Usage:**

- It is used when cutting a **sheet material** (MDF, acrylic, etc.) with a thickness of about 1mm to 10mm in **2D** according to the given 2D drawing.
- In the case of high-power equipment, it is possible to cut metal plates such as steel, but since it is beyond the scope of ME400, only low-power equipment is considered.

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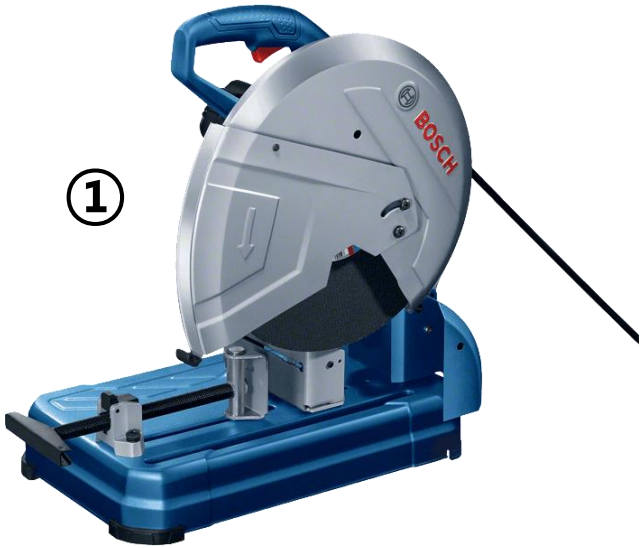

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Other Tools

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Machine Tool Properties: Other Tools

- **Usages:**

1) High Speed Cutter: Non-ferrous metal, Wood, Plastic cutting. (Defending on the saw blade.)
2) Band Saw: Wood, Plastic cutting. (No Steel!)
3) Drilling Machine: Steel, Non-ferrous metal, Wood, Plastic, etc. drilling. (No heat treated steel!)
4) Hand Drill: Non-ferrous metal, Wood, Plastic, etc. drilling.
5) Hand Grinder: Steel grinding, cutting. (Only for steel!)
6) Mini Hand Grinder(DREMEL): Steel grinding, cutting. (Only for steel!)
7) Belt Sander: Almost all materials that can sand sanding. (No sharp metals!)

+ Numbering based on the picture on the previous slide.

Attachment & Tolerance

Attachment & Tolerance: Attachment

- Below are the attaching methods often used in prototyping.

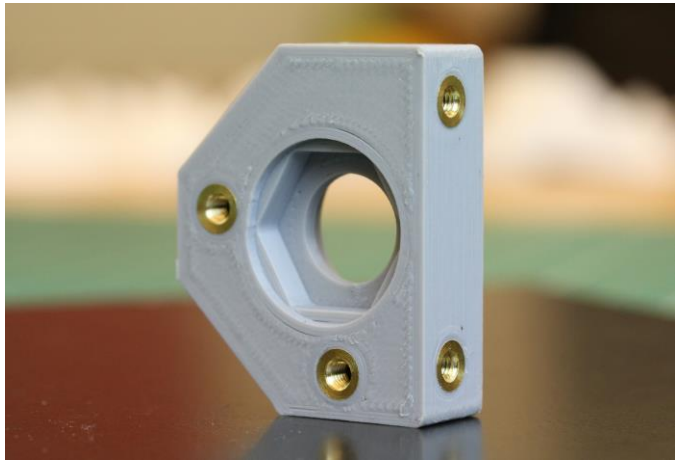
	Pros	Cons
Bolt-Nut	Need simple holes only.	Need wrench during assembly.
Bolt-Tap	No wrench is needed during assembly.	Need to Tap. Need enough depth for Tap hole for conjunction force.
Bolt-*insert or Helicoil*	No wrench is needed during assembly.	Need to insert *insert* it. Need enough depth for it.

When using a nut, a space and path are needed to put a wrench on the nut side from the outside.

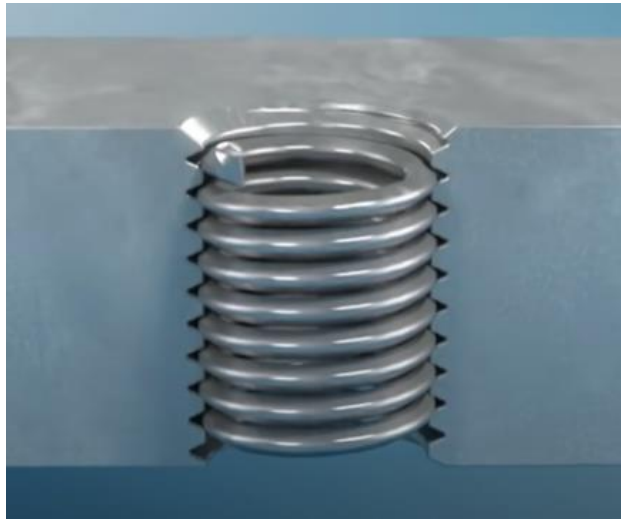
- Glue(Loctite bond, Glue gun, Tape, etc.)
- Press fit

Attachment & Tolerance: Attachment

Brass insert



Helicoil



Tap

Bolt-Nut

Attachment & Tolerance: Tolerance

- Below are recommended tolerance for PLA 3D printed parts with well-maintained FDM 3D printer.

- **Bolt** hole: **(n+0.2~0.5)**Ø for M(n) Tap.

  Ex) 4.0+0.3=4.3Ø hole on drawing for M4 Bolt.

- **Tap** hole: **+0.2Ø +*Tap Drill Chart*** on the right.

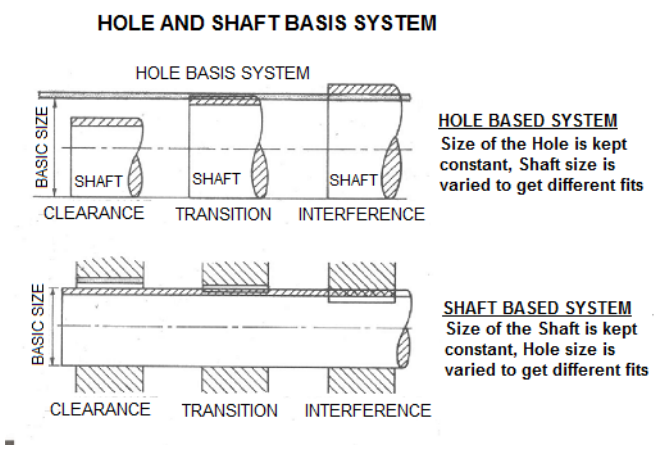
  Ex) 3.3+0.2=3.5Ø hole on drawing for M4 Bolt.

- **Shaft**(Steel) Press fit: Same nominal diameter with the shaft in drawing.

- Between 3D printed parts: **+0.2~+0.5mm** difference in Length or Diameter.

+When a part with a hole is printed by FDM printer, the diameter tends to come out smaller than the size on the drawing. This is due to the diameter of the nozzle.

+Of course, since this is a rough advice, tolerance adjustment in units of 0.1mm may be experimentally necessary.



Metric Coarse Tapping Drill Sizes

Size	Pitch	Drill
M1	0.25	0.75
M2	0.4	1.6
M3	0.5	2.5
M4	0.7	3.3
M5	0.8	4.2
M6	1	5
M7	1	6
M8	1.25	6.75
M9	1.25	7.75
M10	1.5	8.5
M11	1.5	9.5
M12	1.75	10.2
M14	2	12
M16	2	14
M18	2.5	15.5
M20	2.5	17.5
M22	2.5	19.5
M24	3	21

GCodeTutor.com

Electronics Properties

DC motor

Electronics Properties: DC motor

- **Usage:**

- **(Brushed) DC motor**, which is generally used as the main actuator for vehicles.

Main Variables:

- **<u>Weight / Max Torque / Rated Torque / No Load Speed / Voltage / Max Current / Max Power</u>**

In the case of cheap models, the above values may not be provided.

- **Characteristics:**

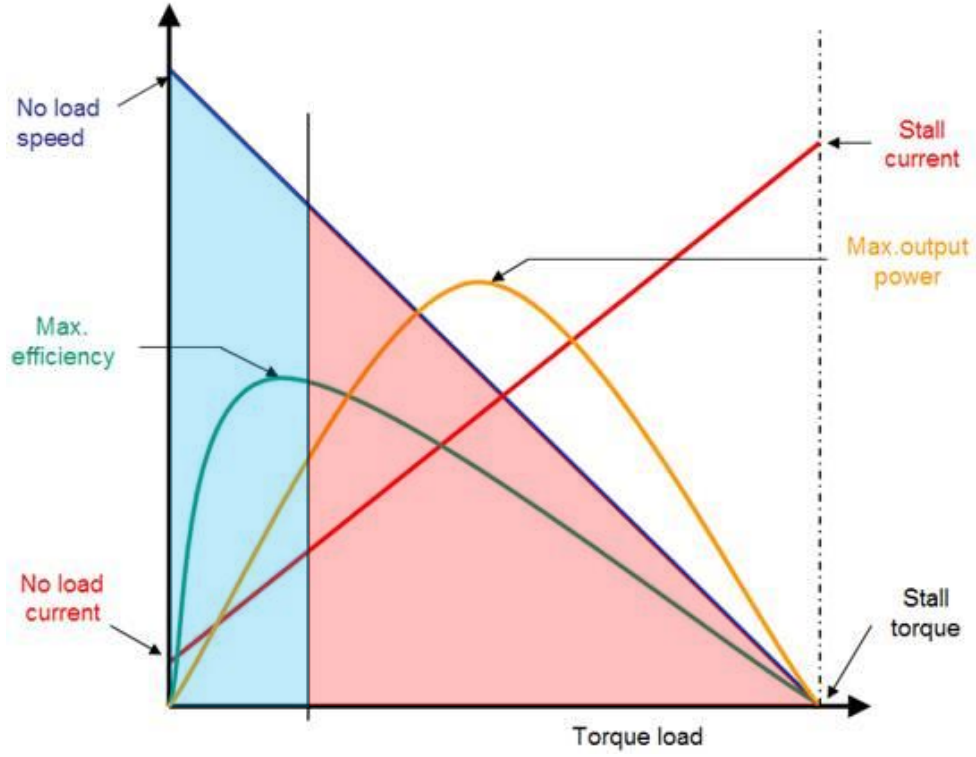
- It is recommended that the motor mount does not contact with plastic parts as it generates considerable heat.
- Motors which have **max power lower than 250W** are recommended. Over 250W, it will be too powerful and dangerous.

Electronics Properties: DC motor

- **Brief Brushed DC Motor Dynamics:**

- The **torque** of a motor is almost linearly proportional to the **current**.$(\tau \propto i)$
- The no load **speed** (RPM) of a motor is almost linearly proportional to its **voltage**.$(\omega \propto V)$
- To prevent *armature* damage due to heat generation, most DC motors have a limit value of **stall current** or **stall torque**. If it exceeds the limit, the motor *armature* will burn and be permanently damaged.

[DC Motor T-N curve]



Electronics Properties: DC motor

- **Motor + Gearhead Selection guide**

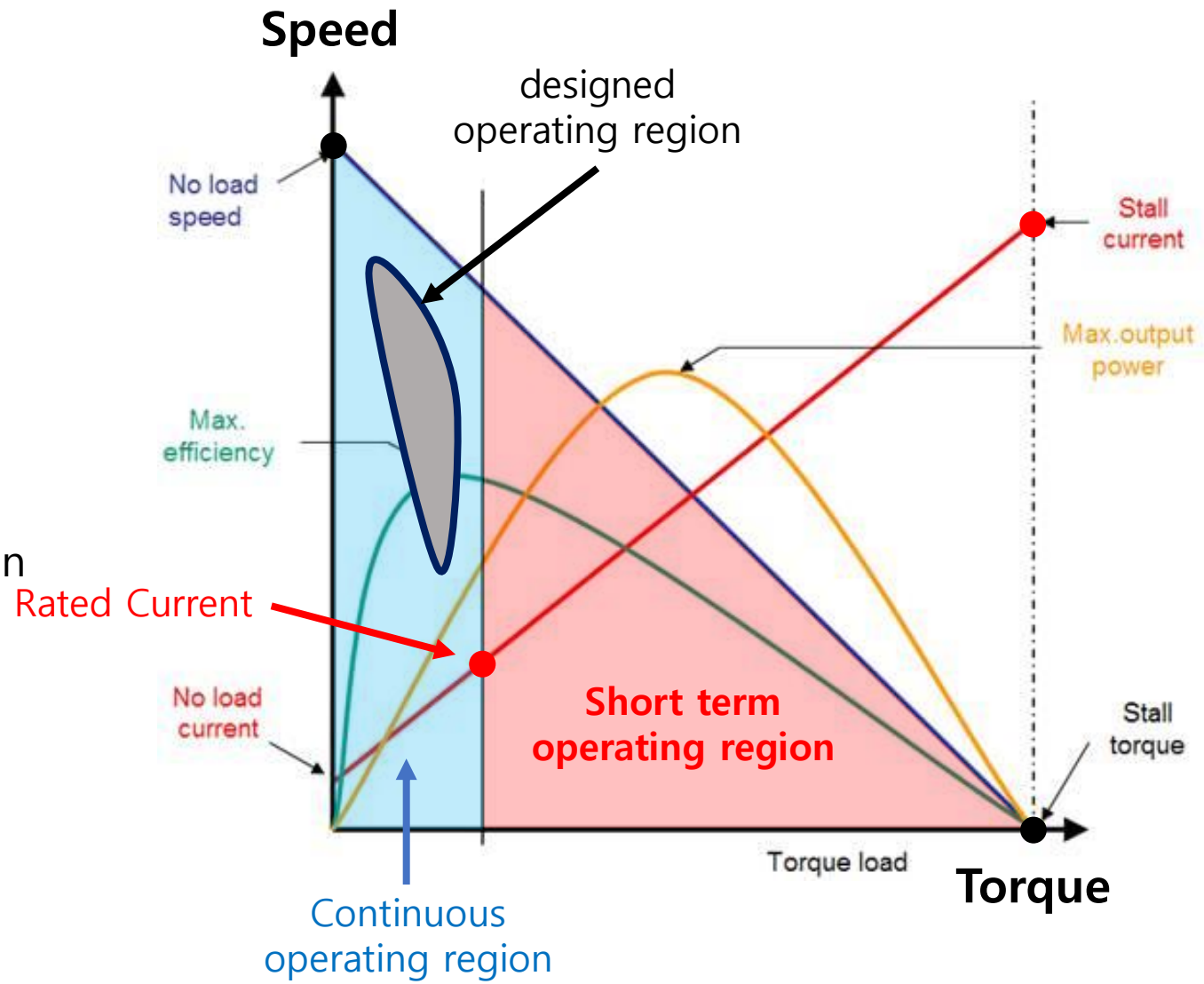
- In the case of cheap models,

  T-N curve may not be provided.

- Ensure that your designed operating region is within the operating region.

- If operated for an extended period within the short-term region, the motor's armature could sustain **damage** due to **heating** caused by current, potentially leading to **permanent damage.**

[DC Motor T-N curve]

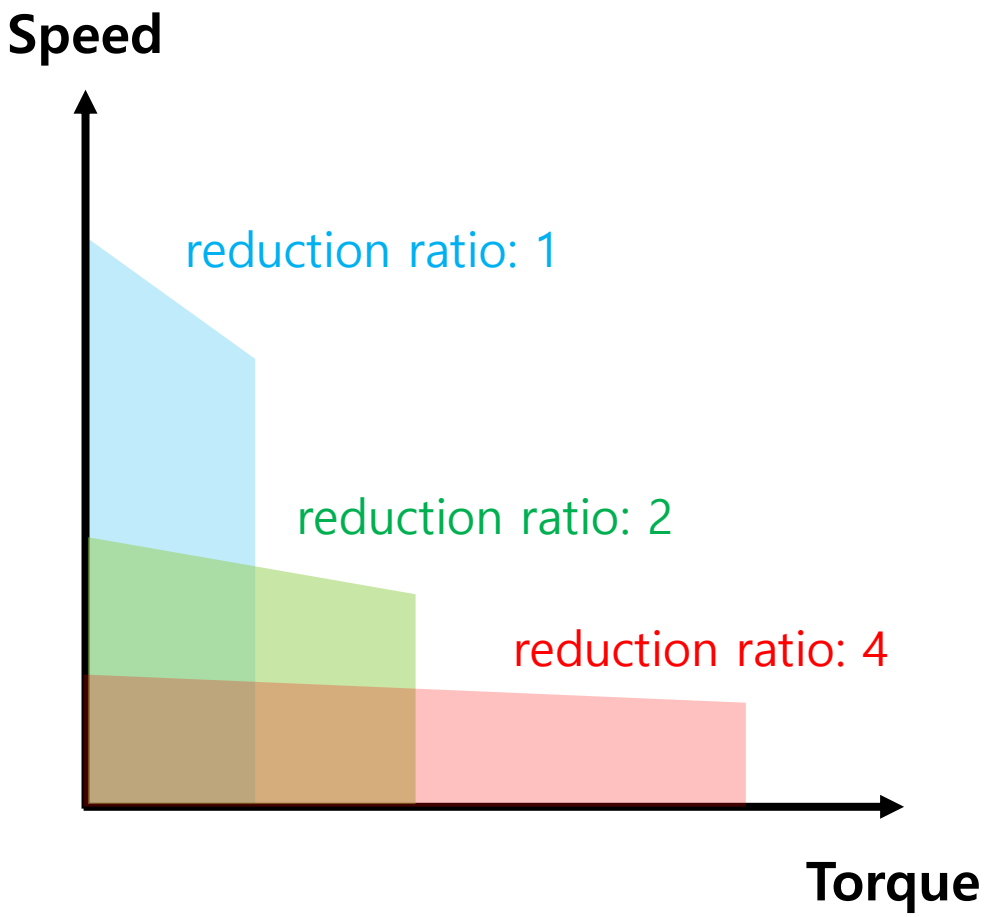


Electronics Properties: DC motor

- **Motor + Gearhead Selection guide**

- Effect of Gearhead.
- torque ←torque * reduction ratio.
- speed ←speed / reduction ratio.

EX)



Electronics Properties: DC motor + Mounting hub

- **Mounting hub & D-cut**

- Used to connect motor shaft and other components.

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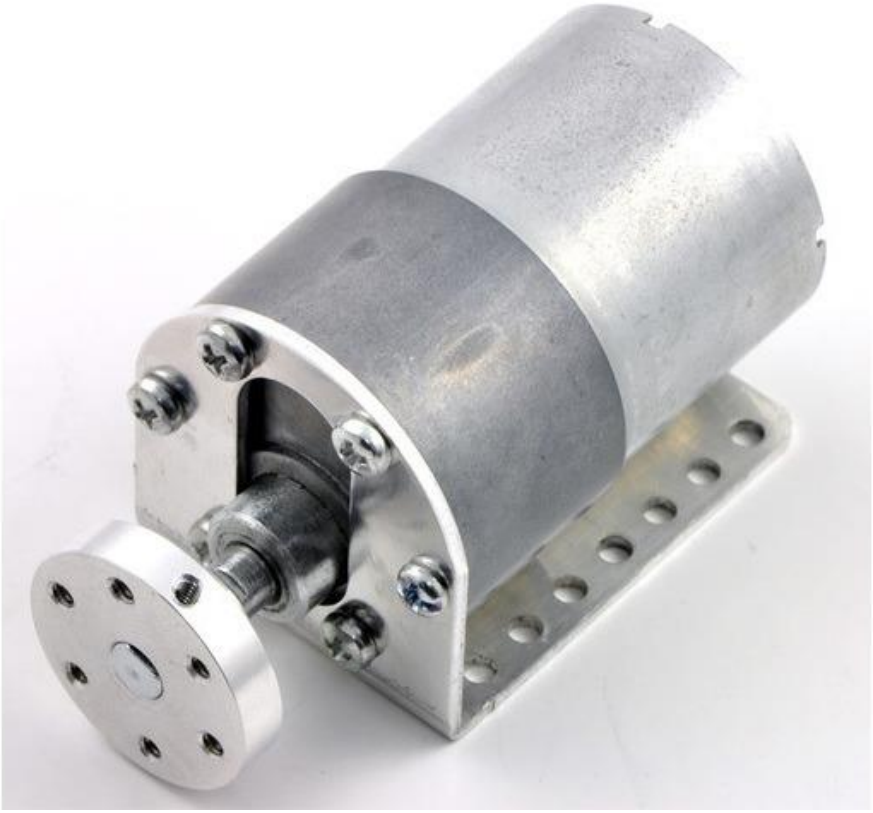
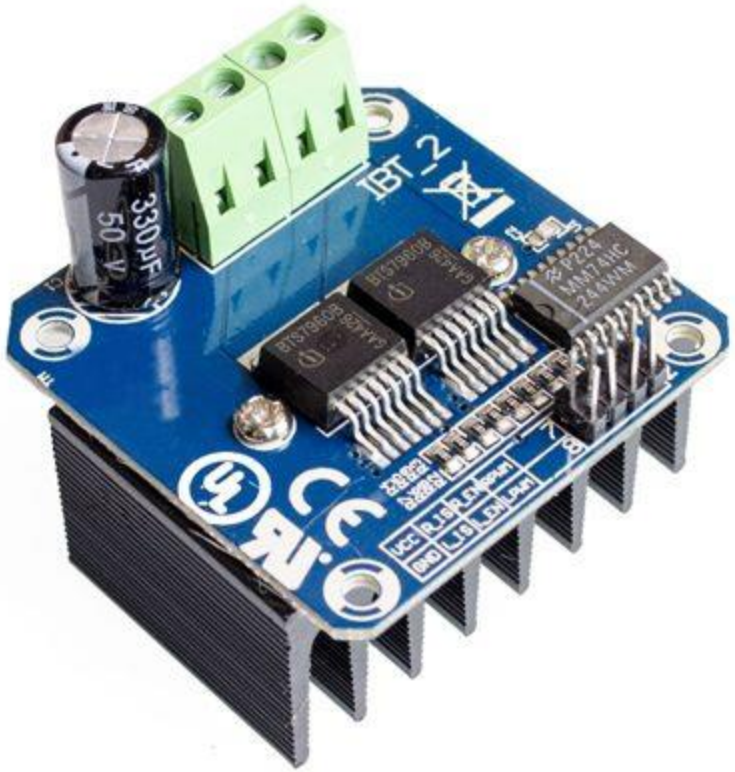
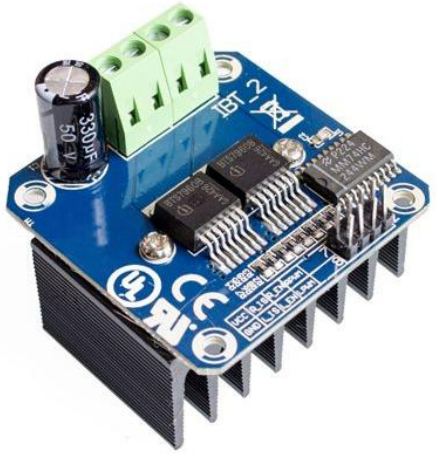




DC motor Driver

Electronics Properties: DC motor Driver



- **Usage:**

- It converts low-current (~20mA) **voltage** signals from the controller into high-current (~50A) **voltage** outputs for the motor.

Main Variables:

- **<u>Supply Voltage / Max Current / Max Power / Weight</u>**

- **Characteristics:**

- It controls the voltage applied to the motor.
- It switches the input voltage (around 20kHz) using the same duty cycle as the PWM signal received as input (20~500Hz).
- Typically, the direction of the motor (voltage) can also be reversed through a separate input signal.

LiPo battery

Electronics Properties: LiPo battery

- **Usage:**

- High current power source.

- **Main Variables:**

- **<u>Weight / Capacity / # of Cells / C-Rating</u>**

- **Characteristics:**

- The method with the **highest discharge current.**
- C-Rating(방전율)[C] x Capacity[Ah] = Max current[A]. There are Continuous and Peak C-rating.
- LiPo battery's nominal voltage per 1 cell is 3.7V, max for 4.2V and min for 3.0V. If a battery is over discharged(Cell voltage<2.7V), the battery is permanently damaged. **DO NOT USE IT**.
- For charging, 1C is recommended (ex. For 1,000mAh battery, charge with 1A setting.)
- For more detailed instruction, visit URL below.

https://www.rogershobbycenter.com/lipoguide
https://m.blog.naver.com/flyfuturek/221171435960

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Li-ion battery

Electronics Properties: Li-ion battery

- **Usage:**

- High energy density power source.

- **Main Variables:**

- **<u>Weight / Capacity / Continuous Current limit</u>**

- **Characteristics:**

- The method with the **highest discharge current.**

- **18650 type** is the most popular.

Kill switch

Electronics Properties: Kill switch

- **Usage:**

- Switch for emergency power cut.

- **Main Variables:**

- **<u>Weight / Max Current</u>**

- **Characteristics:**

- Since the BLDC motor system is very dangerous in case of malfunction as it generates large power, a switch that can quickly cut off the power in an emergency is necessary. While general small switches can withstand less than about **3A** of current, general kill switches can withstand more than **100A** of current.

Servo motor

Electronics Properties: Servo motor

- **Usage:**

- Angle control with motor.

- **Main Variables:**

- **<u>Weight / Stall torque(or peak torque) / No load RPM / Max current / Voltage</u>**

- **Characteristics:**

- Generally they need **6V or 12V** power supply.
- Most analog servos require **PWM** signals, **50Hz and 1 to 2ms** pulse width band.
- Digital servos tend to be more accurate, but requires complicated protocol for signal.
- Due to brushed DC motor in it, it makes voltage **noise**. So Do Not share same Vee(+) line with other complicated electronic parts(ex. Arduino, Raspberry Pi, all sensors).
- Low cost servos tend to show low angular precision & accuracy and can not produce peak torque on the spec.

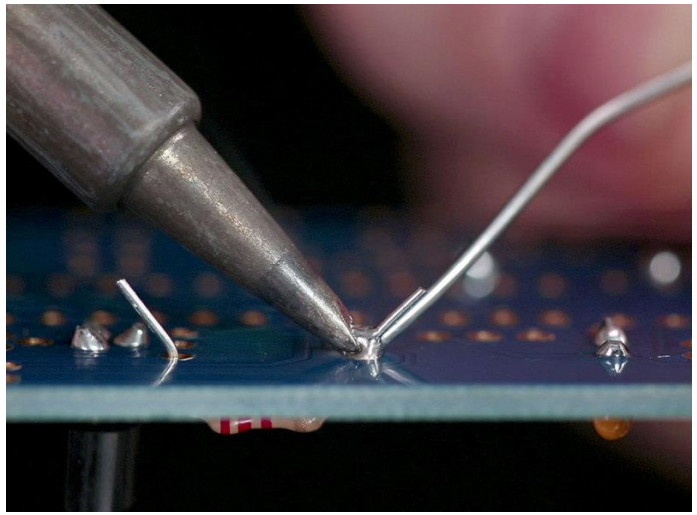
Electronics Properties: Servo motor

- **Widely used model and brands:**

- Micro servo: MG90S(TowerPro)
- Analog servo: HS series(Hitec)
- Digital servo: Dynamixel AX series(Robotiz)

Soldering Tips

Soldering Tips



- **Brief Soldering Tips:**

- The contact area between the object to be melted and the *soldering iron* should be wide so that the object melts quickly. Therefore, it is recommended to touch the object with a small amount of solder on the tip of the *soldering iron.*
- *Soldering iron* tips that become soiled during use can be cleaned by scrubbing them with an iron tip cleaner (Brass sponge).
- If you use *soldering paste or flux* after put it to the tip of the *soldering iron*, the solder will stick more easily to the object you want to put solder on.
- For more detailed procedure, visit URL below.

https://www.makerspaces.com/how-to-solder/
https://kocoafab.cc/tutorial/view/512

Soldering Tips: Wire

- **Brief Soldering Tips:**

- The thickness of the wire is indicated by a *gauge*, and the smaller the number, the thicker it is.
- When connecting wires, after soldering, be sure to use *heat-shrink tubes* to prevent unexpected *short-circuits.*
- When soldering the wires of the battery, be especially careful not to cause *short-circuit.*

- For more detailed instruction, visit URL below.

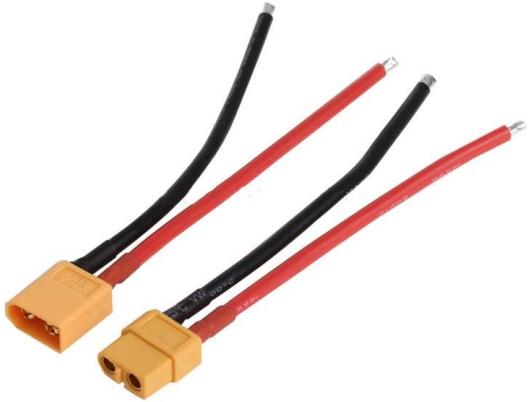
https://www.wikihow.com/Solder-Wires-Together
https://www.youtube.com/watch?v=qxqZJH3SfN4

Soldering Tips: Connector & Shrink Tube

XT series connectors

- **Brief Soldering Tips:**

- When selecting the type of connector, connect the F connector to the power supply side and the M connector to the power consuming side This is because if M connector is connected to the power source, some conductor may enter the connector and the two terminals may be short-circuited.



- For more detailed procedure, visit URL below.

  https://www.youtube.com/watch?v=w0pMjZ-rFQA

Appendix

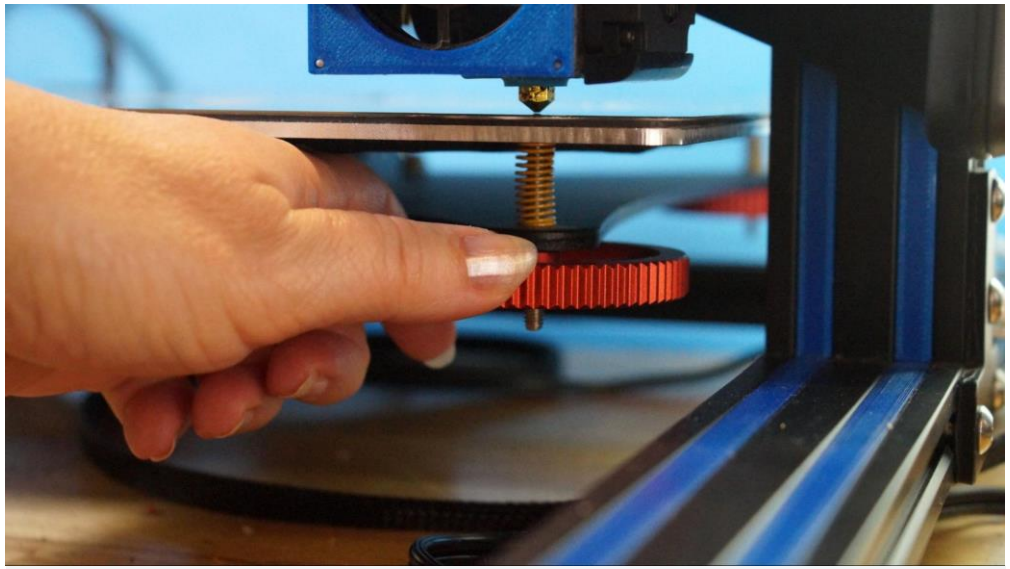
Machine Tool Properties: FDM 3D Printer

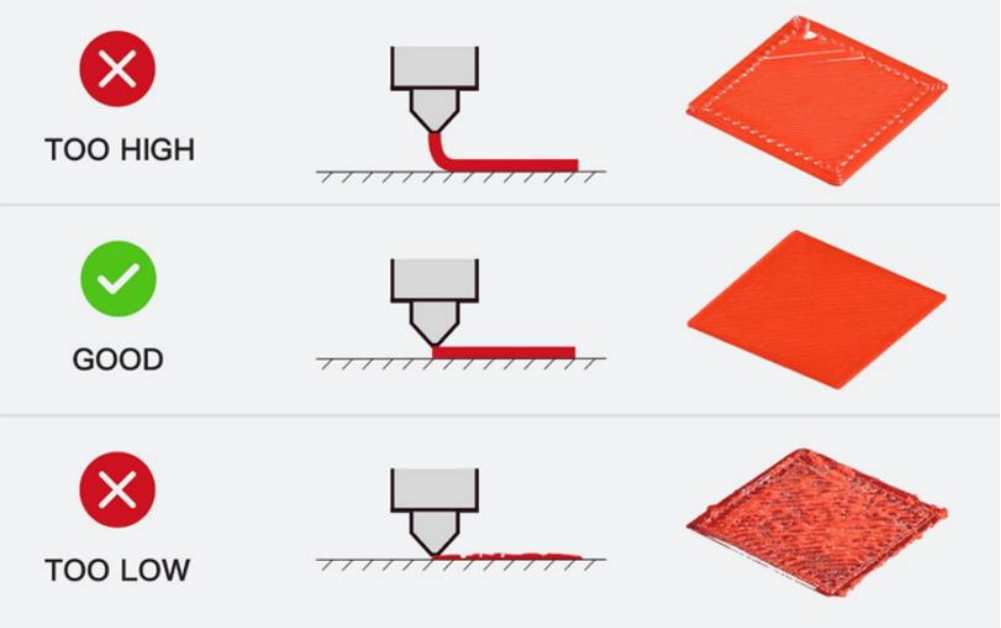
- 3D Printer **Maintenance** Tips:

Tilted leveling:

- If the first layer does not stick to the build plate and falls off (Too far), or if the filament transfer motor generates periodic crackling noise (Too close), The *leveling feet* under the *build plate* should be turned so that the z axis of the build plate and printer is as vertical as possible.
- For the leveling method, refer to the manufacturer's manual of each printer.

+Expensive printers often support *auto leveling.*



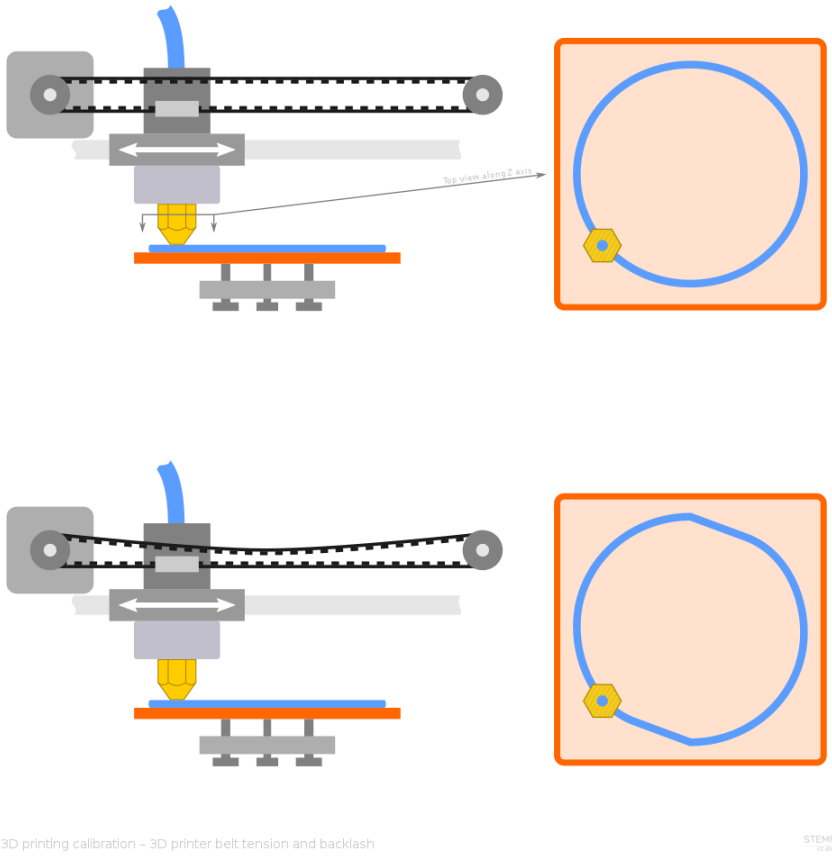


Machine Tool Properties: FDM 3D Printer

- 3D Printer **Maintenance** Tips:

Loose timing belt tension:

- If the printed object as a shape that is pushed smoothly unlike the design, it may be caused by a lowered tension of the timing belt. In this case, it can be solved by increasing the tension of the timing belt.



Normal Loose tension

Machine Tool Properties: FDM 3D Printer

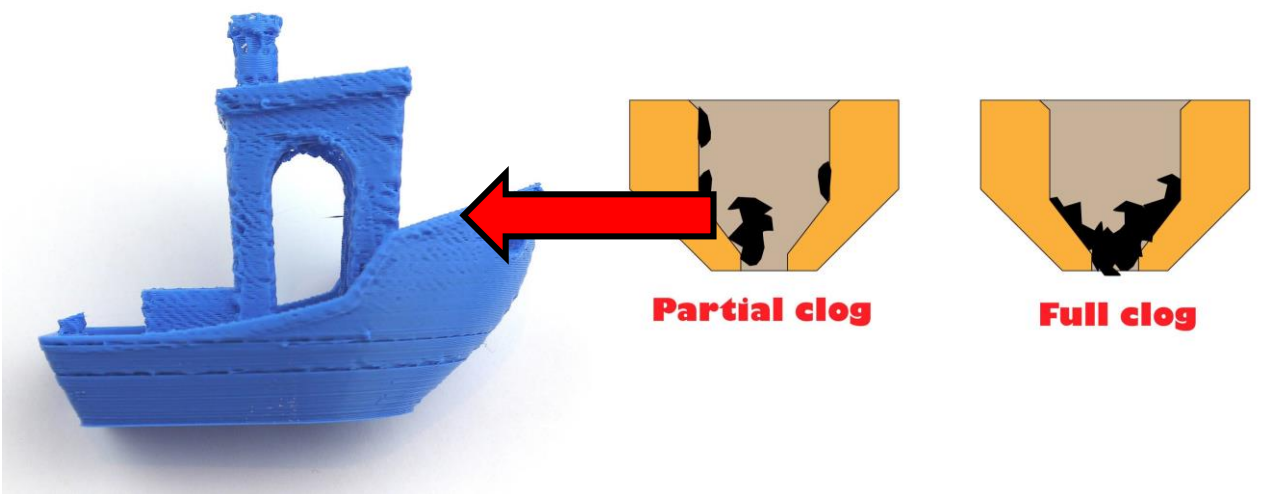
- 3D Printer **Maintenance** Tips:

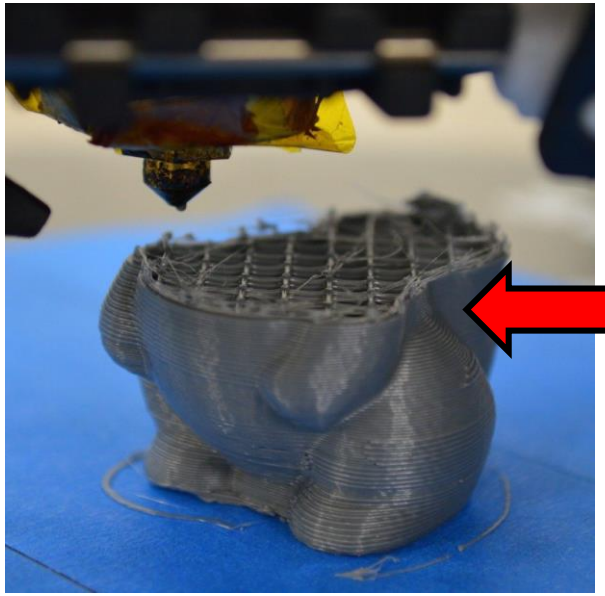
Clogged nozzle:

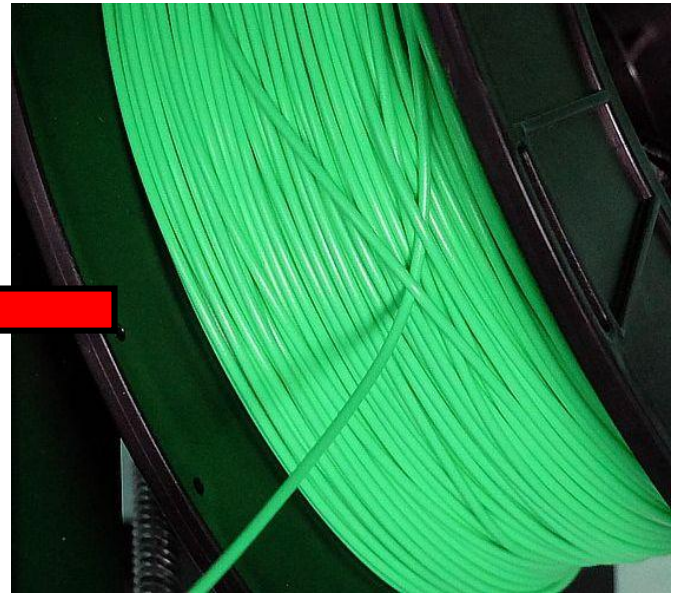
- If the filament does not come out uniformly throughout the printing process and the amount of filament extrusion decreases, a clogged nozzle may be the cause. In this case, the nozzle must be cleaned or replaced.

Tangled filament:

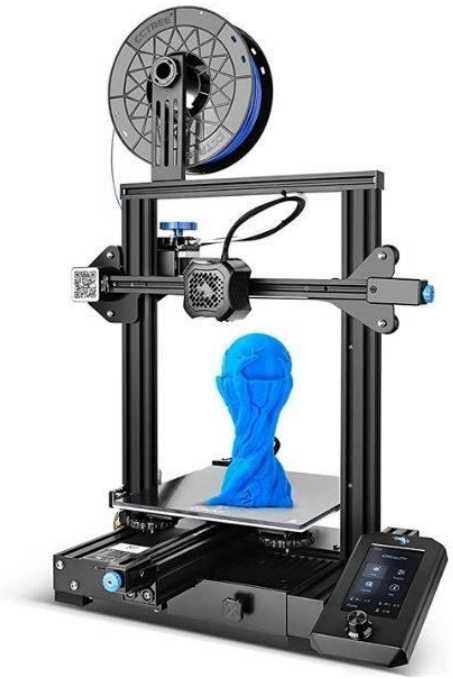
- If extrusion of filament is stopped after the certain layer, it may be due to tangled filaments in the filament roll.







Machine Tool Properties: FDM 3D Printer



- **Safety:**

Although no disease or harmfulness to the human body caused by FDM 3D printers has been clarified yet, a certain amount of **volatile organic compounds and nanoparticles** are generated in the process of heating the plastic filament. Therefore, when using an FDM 3D printer, it is most recommended to use it in a place equipped with **exhaust facilities**. If it is difficult, it is recommended to keep the printer away from the place where people are usually present and to **ventilate** the room frequently. In the case of low-cost open type printers, it is also recommended to cover them with a large paper box to block them from outside air.

END